

Solution Requirements

Date	15/07/2026
Team ID	SWTID-2026-3095
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Description	Priority
1	User Form Input	Web interface accepting applicant demographic and financial parameters (income, employment, education, family status, housing, occupation, etc.).	High
2	Data Preprocessing	Automatic categorical encoding, remapping of rare occupation values to "Other" (matching training-time grouping), and derived feature calculation (income per person, log income, contact score).	High
3	Column Alignment	Reindex the encoded input row to the exact column order used at training time (model_columns.pkl), filling any missing dummy columns with 0.	High
4	Model Inference	Run the tuned XGBoost classifier on the aligned feature vector to produce a high-risk prediction and probability score.	High
5	Result Presentation	Render a clear results page showing Approved / Needs Review status and the underlying risk probability percentage.	High
6	Error Handling	Validate required form inputs and gracefully handle unrecognized categorical values (e.g. an occupation not seen during training) to prevent server crashes.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Low latency response time for a single prediction request.	< 1.5 seconds per prediction
2	Model Discrimination	Meaningful separation between high-risk and low-risk applicants despite severe class imbalance (1.69% positive rate in training data).	> 0.70 ROC-AUC on held-out test set
3	Security	Input validation and safe handling of unexpected or unseen categorical values.	Sanitized inputs; no crash on bad or unseen values
4	Usability	Clean, responsive form and result UI usable across screen sizes.	User-friendly CSS layout, mobile-responsive
5	Maintainability	Model artifacts (best_model.pkl, model_columns.pkl, categorical_value_map.pkl) are versioned and reproducible from the notebook.	Notebook reproducible via Restart Kernel & Run All